

LQR

$\dot{x} = Ax + Bu$ , how to choose  $u$  so that pole stays upright? LQR:  $u = -Kx$

$K$  is chosen to minimize  $J = \int_0^\infty (x^T Q x + u^T R u) dt$

Q: How much we care about state error

R: ————— || ————— using large control force ( $\omega$ )

Initially:

$$Q = \begin{bmatrix} 1.0 & 0 & 0 & 0 \\ 0 & 0.1 & 0 & 0 \\ 0 & 0 & 100 & 0 \\ 0 & 0 & 0 & 1.0 \end{bmatrix} \Rightarrow \text{pole angle matters a lot.}$$

$R = 0.1 \rightarrow$  Don't necessarily want wage forces.

We check the controllability and see that

$$\text{Rank}(C) = 4$$